

Math

As well as storing numbers in variables (see pp.50–51), Scratch can be used to carry out all sorts of calculations using the “Operator” blocks.

SEE ALSO

◀ 50–51 Variables

Math 112–113 ▶
in Python

Doing sums

There are four “Operator” blocks that can be used to do simple calculations. These are addition, subtraction, multiplication, and division.

$$(7) + (22)$$

△ Addition

The “+” block adds the two numbers in the block together.

$$(64) - (28)$$

△ Subtraction

The “–” block subtracts the second number from the first.

The “think” block is used here to print the result

think

$$(2) + (5)$$

△ Printing results

Drag a “think” block into the scripts area and drop a “+” block inside it. Now add two numbers together and watch your sprite think the answer.

$$(11) * (10)$$

△ Multiplication

Computers use the “*” symbol for multiplication, because “x” looks like a letter.

$$(120) / (4)$$

△ Division

There’s no division sign on the keyboard, so Scratch uses the “/” symbol instead.

Results in a variable

For more complex calculations, such as fixing the sale price of an item, instead of just using numbers you can use the value of a variable in a sum. The result can be stored in a variable too.

Variables are useful if you want to repeat the same sum with different values.



1 Create variables

Go to the “Data” section of the blocks palette and create two variables—“sale price” and “price”.

2 Set the price

Select the “set price” block and fix the price of an item to 50.

set price ▼ to 50

Use the drop-down menu to select “price”

3 Calculate the sale price

Use this script to calculate half the price of an item and set it as the sale price.

set sale price ▼ to

price

/ (2)

Drag the “price” variable into the window to divide the price by 2

Add the “/” block inside the “set sale price” block